

ASSIGNMENT-11

NAME: SRAVANI ANUMALA

ROLL NO: 2303A510G7

BATCH: 30

CODE:

```
class FabricNetwork:
```

```
    def __init__(self):
```

```
        self.ledger = {}
```

```
    # Create Asset
```

```
    def create_asset(self, asset_id, owner):
```

```
        self.ledger[asset_id] = {"ID": asset_id, "Owner": owner}
```

```
        print("Asset Created (Fabric) ✅:", self.ledger[asset_id])
```

```
    # View Ledger
```

```
    def view_assets(self):
```

```
        print("\nFabric Ledger State 📄:")
```

```
        for asset in self.ledger.values():
```

```
            print(asset)
```

```
    # Transfer Ownership
```

```
    def transfer_asset(self, asset_id, new_owner):
```

```
        if asset_id in self.ledger:
```

```
            self.ledger[asset_id]["Owner"] = new_owner
```

```
            print("Ownership Transferred (Fabric) ✅:", self.ledger[asset_id])
```

```
        else:
```

```
            print("Asset not found ❌")
```

```
# Simulating Ethereum (just for comparison)
```

```
class EthereumNetwork:
```

```
    def transfer(self, sender, receiver, amount):
```

```
        gas_fee = 0.01
```

```
        print("\nEthereum Transaction 💸 ")
```

```
        print("From:", sender, "To:", receiver)
```

```
        print("Amount:", amount, "ETH")
```

```
        print("Gas Fee:", gas_fee, "ETH ❌ (High)")
```

```
        print("Status: Slower Confirmation ⏳ ")
```

```
# Execution
```

```
fabric = FabricNetwork()
```

```
eth = EthereumNetwork()
```

```
# Fabric operations
```

```
fabric.create_asset("A1", "Sravani")
```

```
fabric.create_asset("A2", "Sunny")
```

```
fabric.view_assets()
```

```
fabric.transfer_asset("A1", "Sunny")
```

```
fabric.view_assets()
```

```
# Ethereum comparison
```

```
eth.transfer("Sravani", "Sunny", 5)
```

